CS228 : Logic for Computer Science

Module 5: Temporal Logic

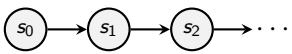
Lecture 23: Linear-time Temporal Logic (LTL)

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Recap: Why temporal logic?

Linear-time $\rightarrow$ LTL

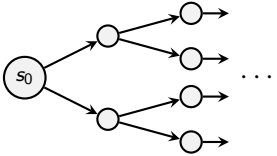


look along *one path*:
one future at each moment



Loki: the Sacred Timeline,
the TVA prunes every branch

Branching-time $\rightarrow$ CTL



view time as a tree, rooted at present:
many possible futures

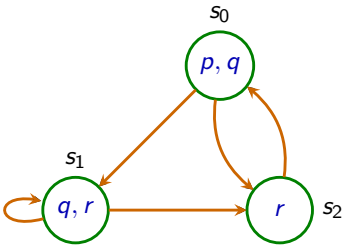


Doctor Strange: millions of futures,
only one where we win

Topic 23.1

Transition Systems

Transition systems as models



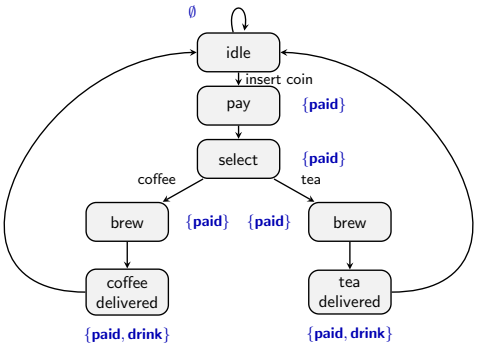
A directed graph: each node (state) contains the atomic propositions true in that state

Exercise 23.1

1. Let $AP = \{p, q, r\}$. List all elements of $\mathcal{P}(AP)$.
2. Write the formal description of M in the diagram.
3. Can M have a deadlocked state, a state from which there is no outgoing transition?
4. If we remove the deadlock condition from the definition, does it matter?

A real transition system: a vending machine

"A vending machine takes a coin, brews coffee or tea, then delivers it."



Atomic propositions $AP = \{\textbf{paid}, \textbf{drink}\}$: the user has paid; a drink is delivered.

Quick questions

1. Which states can follow *select*? two different states, both called *brew*: one for coffee, one for tea!
2. What is the label of *idle*? $\emptyset$: no atomic proposition is true there, a legal label.

Which properties of this machine would we like to state?

"A drink is never delivered before it is paid for."

"Once it is paid for, a drink is eventually delivered."

"Once a drink is delivered, user has to pay before getting a drink again."

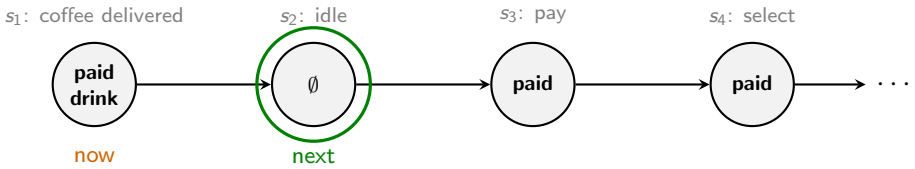
Each one talks about how states follow one another over time: how can we write this in logic?

Topic 23.2

Temporal Connectives

Time by example: next (X)

"If a drink is delivered now, the user has to pay again in the *next* state."



drink $\rightarrow X \neg$**paid**

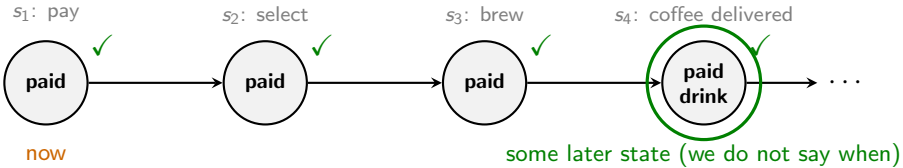
$X\varphi$: φ holds in the "neXt state"

Exercise 23.2 Does the above path, at the current state s_1 (marked *now*), satisfy

1. **drink** $\rightarrow X$ **paid**? No: **paid** does not hold at s_2.
2. **paid** $\rightarrow X$ **paid**? No: **paid** holds at s_1 but not at s_2.
3. **drink** $\rightarrow X\,X$ **paid**? Yes: **paid** holds at s_3.

Time by example: eventually (F)

"The machine *eventually* delivers a drink (**drink**)."



F **drink**

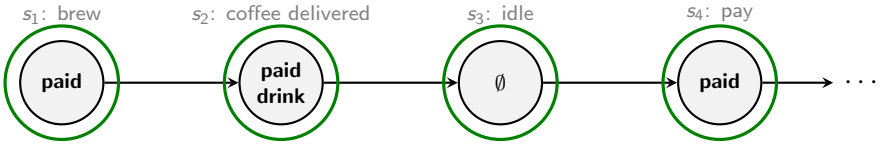
$F\varphi$: φ holds in "some Future state" (possibly the current one)

Exercise 23.3

1. At which states of this path does F **drink** hold? At $s_1, \ldots, s_4$: every state upto and including drink.
2. Does F **drink** hold if **drink** already holds in the current state? Yes: the current state counts.
3. If **paid** holds now, is a drink eventually delivered on this path?
   That is, does **paid** $\rightarrow F$ **drink** hold on this path? Yes: **paid** holds at s_1, and so does F **drink**.

Time by example: always (G)

"Whenever a drink is delivered (**drink**), it has been paid for (**paid**)."



drink $\rightarrow$ **paid** holds in every state, from now on

$$G\,(\textbf{drink} \rightarrow \textbf{paid})$$

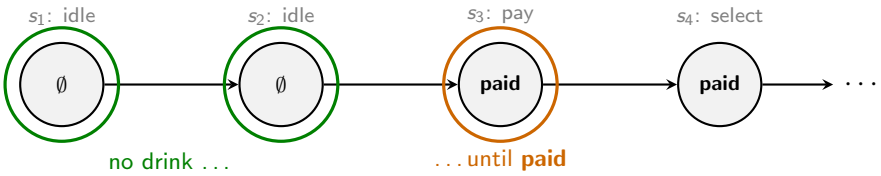
$G\varphi$: φ holds "Globally": in the current state and all later ones

Exercise 23.4

1. Does G **paid** hold on this path? No: **paid** fails at s_3 (idle).
2. Does $G(\textbf{paid} \rightarrow \textbf{drink})$ hold on this path? No: **paid** holds at s_1 but **drink** does not.
3. What would a path look like on which $G(\textbf{drink} \rightarrow \textbf{paid})$ fails? A state where **drink** holds but **paid** does not: a free drink!
4. Write in LTL: "every time the user has paid, a drink is eventually delivered". $G(\textbf{paid} \rightarrow F\,\textbf{drink})$

Time by example: until (U)

"No drink is delivered *until* the user has paid."



¬drink U paid

$\varphi\, U\, \psi$: φ holds "Until" ψ holds (and ψ does hold)

Exercise 23.5

1. Does ¬**drink** U **paid** hold at s_2? Yes: **paid** holds at s_3, and ¬**drink** holds at s_2.
2. Does it hold at s_3, where **paid** already holds? Yes: ψ holds immediately, so φ is not needed.
3. If the user never pays (idle forever), does it hold? No: ψ must eventually hold.
4. Which weaker property says "no drink unless the user pays", allowing that the user never pays? ¬**drink** W **paid**, the weak until (as we will see later).

Topic 23.3

LTL Syntax

Linear-time Temporal Logic (LTL): Syntax

The syntax of LTL in Backus–Naur form (BNF):

$$\varphi ::= \top \mid \bot \mid p \mid (\neg\varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \to \varphi)$$
$$\mid (X\varphi) \mid (F\varphi) \mid (G\varphi) \mid (\varphi \; U \; \varphi)$$

where p is an atomic proposition from set AP.

- $\top$, $\bot$ and the usual propositional connectives $\neg, \wedge, \vee, \to$ are as in propositional logic.
- In addition we have temporal connectives:
  - $X\varphi$: φ holds in the "neXt state"
  - $F\varphi$: φ holds in "some Future state" (possibly the current one)
  - $G\varphi$: φ holds "Globally": in the current state and all later ones
  - $\varphi \, U \, \psi$: φ holds "Until" ψ holds (and ψ does hold)
- Brackets can be dropped using binding rules ("$>$" reads "binds more tightly than"):

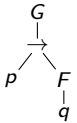
$$\text{Unary } (\neg, X, F, G) \;>\; U \;>\; \vee, \wedge \;>\; \to$$

Parsing LTL formulae

$$\varphi ::= \top \mid \bot \mid p \mid (\neg\varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi)$$
$$\mid (X\varphi) \mid (F\varphi) \mid (G\varphi) \mid (\varphi U \varphi)$$

Examples

1. $G\ (p \rightarrow F\ q)$



2. $F\ (p \rightarrow G\ r) \vee \neg q\ U\ p$
3. $F\ p \rightarrow G\ r \vee \neg q\ U\ p$

Parse trees

- As usual we can write Parse trees for the formulas
- A subformula of an LTL formula φ is any formula ψ whose parse tree is a subtree of φ's parse tree.

Exercise 23.6

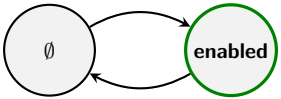
1. Which of these are LTL formulas?
   - 1.1 $(p\ G\ q)$
   - 1.2 $(p\ U\ q)\ U\ r \rightarrow\ F\ p$
   - 1.3 $(p \wedge q) \rightarrow (U\ r) \vee G\ p$
2. Write the parse trees of Examples 2 and 3 and of the formula in part 1 that is an LTL formula.
3. Write all the subformulas of Example 1 above.

Putting it together: writing specifications

What can you express in this logic?

Consider atomic propositions: **req**, **grant**, **enabled**, **running**.

1. Response: In any state, if there is a **request** (of some resource), then it will be eventually **granted**. $G(\textbf{req} \rightarrow F\ \textbf{grant})$
2. Some process is **enabled** infinitely often in every computation path. $G\ F\ \textbf{enabled}$
3. If a process is **enabled** infinitely often, then it **runs** infinitely often. **Exercise 23.7!**



every path keeps coming back to **enabled**

Exercise 23.8 Consider the specification: An upwards moving lift at second floor does not change its direction if someone has pressed button for 5th floor. Come up with atomic propositions and write an LTL formula for this spec.

safety: nothing bad ever happens, e.g., $G(\textbf{drink} \rightarrow \textbf{paid})$

liveness: something good eventually happens, e.g., $F\ \textbf{drink}$

Topic 23.4

LTL Semantics

Semantics of LTL

$$\varphi ::= \top \mid \bot \mid p \mid (\neg\varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi)$$
$$\mid (X\varphi) \mid (F\varphi) \mid (G\varphi) \mid (\varphi U \varphi)$$

- A path in $M = (S, \rightarrow, L)$ is an infinite sequence of states $s_1, s_2, \ldots$ in S s.t., $\forall i \geq 1, s_i \rightarrow s_{i+1}$. Write $\pi = s_1 \rightarrow s_2 \rightarrow \ldots$ and denote by π^i the suffix of π starting at s_i.

Given LTL formula φ, model $M = (S, \rightarrow, L)$ and path $\pi = s_1 \rightarrow \ldots$ in M, the satisfaction relation $\pi \models \varphi$, is defined inductively:

1. $\pi \models \top, \pi \not\models \bot$
2. $\pi \models p$ iff $p \in L(s_1)$
3. $\neg, \wedge, \vee, \rightarrow$ are as usual!
4. $\pi \models X\ \varphi$ iff $\pi^2 \models \varphi$.
5. $\pi \models F\ \varphi$ iff $\exists i \geq 1$, s.t. $\pi^i \models \varphi$.
6. $\pi \models G\ \varphi$ iff $\forall i \geq 1$, $\pi^i \models \varphi$.
7. $\pi \models \varphi\ U\ \psi$ iff $\exists i \geq 1$, s.t. $\pi^i \models \psi$ and $\forall j \in \{1, \ldots, i-1\}$, $\pi^j \models \varphi$.

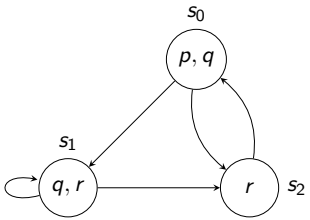
Exercise 23.9 If $\pi = s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \cdots$, what is π^3? Unfold $\pi \models X\,X\,p$ by the definition. $\pi^3 = s_3 \rightarrow s_4 \rightarrow \cdots$; $\pi \models X\,X\,p$ iff $\pi^2 \models X\,p$ iff $\pi^3 \models p$.

Semantics of LTL continued

- We defined a satisfaction relation between paths and LTL formulas.
- But to verify systems, we would like to say the whole model satisfies a formula.

Given LTL formula φ, model $M = (S, \to, L)$ and $s \in S$, we define $M, s \models \varphi$ if, for every path π of M starting from s, we have $\pi \models \varphi$.

Let's see an example.



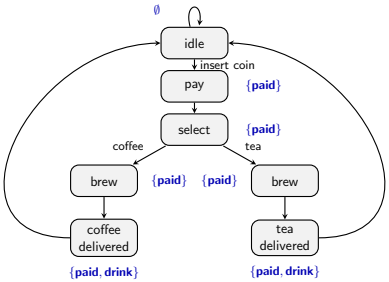
Consider M (left).

1. $M, s_0 \models (p \wedge q)$ ✓
2. $M, s_0 \models \neg r$ ✓
3. $M, s_0 \models X\ r$
4. $M, s_0 \models X(q \wedge r)$ ×
5. $M, s_0 \models G\neg(p \wedge r)$

Which of the following hold?

6. $M, s_2 \models G\ r$
7. $\forall s, M, s \models (F(\neg q \wedge r) \to F\ G\ r)$
8. $M, s_0 \models G\ F\ p$ ×
9. $M, s_0 \models (G\ F\ p \to G\ F\ r)$

Satisfaction on the vending machine



Exercise 23.10 Which of these hold in the vending machine M (with the labels shown)?

1. $M, idle \models F\,\textbf{drink}$? No: the path idle, idle, idle, ... never delivers a drink.
2. $M, pay \models F\,\textbf{drink}$? Yes: every path from pay reaches a delivered state.
3. $M, idle \models G(\textbf{drink} \rightarrow \textbf{paid})$? Yes: **drink** holds only in the delivered states, and they have **paid**.
4. $M, idle \models (\neg\textbf{drink}\ U\ \textbf{paid})$? No: on the path idle, idle, ..., **paid** never holds. (The weak until W would hold.)